nature energy

ARTICLES

https://doi.org/10.1038/s41560-021-00922-6

Check for updates

Understanding environmental trade-offs and resource demand of direct air capture technologies through comparative life-cycle assessment

Kavya Madhu[1]✉, Stefan Pauliuk[1], Sumukha Dhathri[1] and Felix Creutzig[2,3]

Direct air capture (DAC) technologies remove carbon dioxide (CO_2) from ambient air through chemical sorbents. Their scale-up is a backstop in many climate policy scenarios, but their environmental implications are debated. Here we present a comparative life-cycle assessment of two main DAC technologies coupled with carbon storage: temperature swing adsorption (TSA) and high-temperature aqueous solution (HT-Aq) DAC. Our results show that TSA DAC outperforms HT-Aq DAC by a factor of 1.3–10 in all environmental impact categories studied. With a low-carbon energy supply, HT-Aq and TSA DAC have a net carbon removal of up to 73% and 86% per ton of CO_2 captured and stored. For the same climate change mitigation effect, TSA DAC needs about as much renewable energy and land occupation as a switch from gasoline to electric vehicles, but with approximately five times higher material consumption. Input requirements for chemical absorbents do not limit DAC scale-up.

There is a considerable risk that ambitious technology-based transformation across all industrial and end-use sectors will be partly infeasible or happen too late to bring global warming to a halt[1,2]. Atmospheric carbon dioxide (CO_2) removal at an industrial scale may be needed as a so-called backstop[3,4] or emergency[5] technology as demonstrated in prominent mitigation scenarios[2,6–8]. Direct air carbon capture and sequestration (DACCS) refers to technologies that capture and sequester atmospheric CO_2 via cyclic processes involving reversible chemical reactions. In the direct air capture (DAC) step, CO_2 is first bound to a sorbent and then desorbed into a concentrated stream, while the sorbent is regenerated for further cycles through one or more chemical reactions[9]. On the basis of the sorbent and regeneration method used, DAC technologies are classified into high-temperature aqueous solution (HT-Aq), low-temperature solid sorbent utilizing temperature vacuum swing adsorption (TSA) and low-temperature solid sorbent utilizing moisture swing adsorption[10]. In the carbon storage step, the sequestered pure CO_2 stream is compressed, transported and pumped into permanent geological storage.

Capturing and sequestering CO_2 via DAC is an electricity-intensive and heat-intensive process[10–14]. For HT-Aq DAC, high temperature (850–900 °C) is required for the regeneration of the sorbent[10,15], and energy requirements can range from 5.1 to 8.1 GJ per ton CO_2 captured[10,16]. For the regeneration of the sorbent in TSA DAC, low temperature (80–120 °C) is sufficient, and waste heat, heat pumps or electric heaters can therefore be used. Approximately 5–7.5 GJ of heat are required per ton CO_2 captured[10]. In addition to the substantial operational energy demand, DAC requires materials in assets and for its operation, including amine-rich sorbents for TSA DAC, sodium hydroxide (NaOH) or potassium hydroxide (KOH), pure oxygen[11,12,15] and calcium carbonate ($CaCO_3$) for HT-Aq DAC[15] as well as water, land and waste treatment services.

There is limited publicly available information on crucial DAC process parameters and a lack of comparative assessments of main DAC pathways. Also, there is considerable uncertainty regarding the technology learning and efficiency-of-scale potential of DAC. As a consequence, the future resource requirements and potential environmental impacts of DAC and how they compare with those of other pathways, such as bioenergy combined with carbon capture and storage (BECCS), are highly uncertain and a matter of debate[16,17]. This lack of knowledge directly translates into widely varying cost estimates from US$11 to US$1,000 per ton CO_2 captured[3,10,11,15]. In addition to cost variability, unavailable and uncertain long-term geological storage makes it impossible at this stage to robustly design business models, policy guidance and support for a large-scale roll-out of DAC.

Here we provide a detailed quantitative systems analysis of this emerging technology in the form of a life-cycle assessment (LCA) of different DAC types. We compile the relevant process data, compare the environmental performance of different technologies, identify crucial knowledge gaps and simulate the impact of different system changes (for example, low-carbon energy (LCE) supply) and technology learning scenarios. As DAC technologies are modelled to remove CO_2 at gigaton scale from the atmosphere, we also compute a detailed estimate of chemical and material resource demand, an important ingredient for sustainability assessments of negative emission technologies.

LCA of DAC technologies

De Jonge et al.[18] calculated the life-cycle carbon capture efficiency (CCE)—that is, net CO_2 captured (see Methods for a proper definition)—of HT-Aq DAC. Their CCE results vary from 10% to 90%, depending on technology assumptions but do not include the process equipment required for the sorbent regeneration. Liu et al.[19] studied a combination of HT-Aq DAC with a Fischer-Tropsch

[1]Faculty of Environment and Natural Resources, University of Freiburg, Freiburg, Germany. [2]Mercator Research Institute on Global Commons and Climate Change, Berlin, Germany. [3]Sustainability Economics of Human Settlements, Technische Universität Berlin, Berlin, Germany. ✉e-mail: kavya.madhu@indecol.uni-freiburg.de

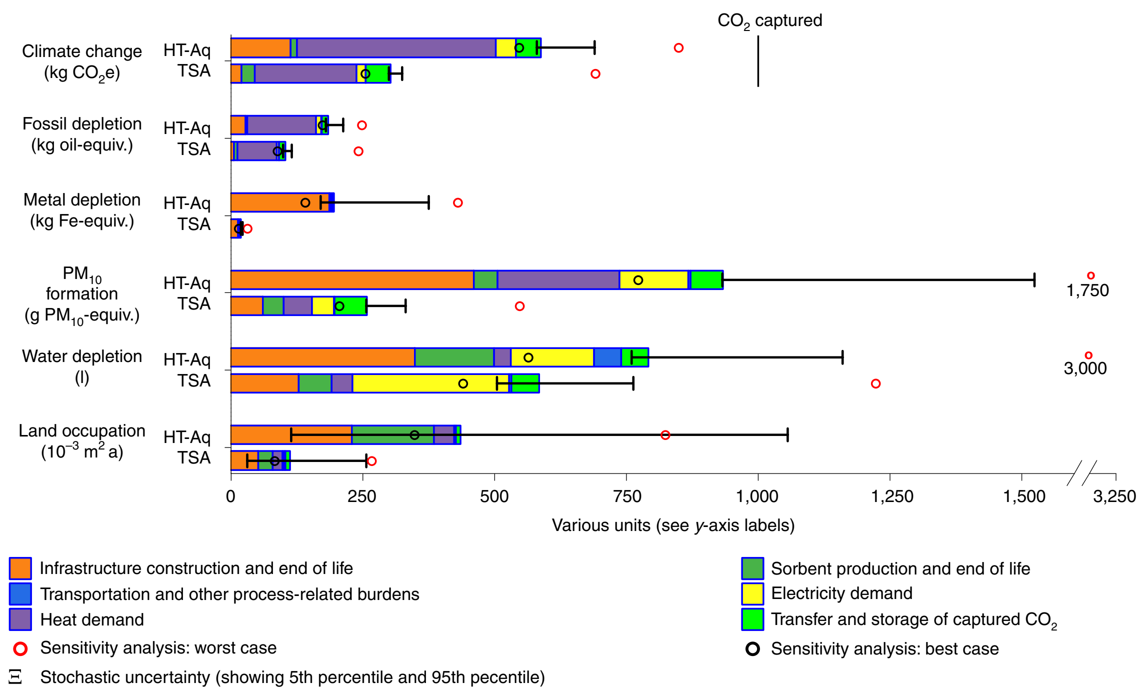

Fig. 1 | Comparative environmental analysis of the reference case (current technology) for HT-Aq and TSA DAC technologies, respectively. The functional unit is 1t of atmospheric CO_2 captured. The variation in the results arises from the parameter changes summarized in Tables 1 and 2. The captured CO_2 (1,000 kg) shown in the figure needs to be subtracted from the climate change bars to obtain the life-cycle climate impact of the DAC technologies. Stochastic uncertainty was determined by conducting a Monte Carlo simulation. Oil-equiv., Fe-equiv. and PM_{10}-equiv. refer to oil-equivalent, iron-equivalent and PM_{10}-equivalent, respectively. The *x*-axis break is from 1,600 to 3,200.

synthesis to produce liquid fuels. Using an aggregated economic input–output database and including the DAC infrastructure, they estimated that HT-Aq DAC emits 1.1–1.5 kg CO_2 per kg CO_2 captured. Their system was found to have 50–90% CCE when the climate benefit of utilizing synthesized fuel instead of conventional fuel was considered. Deutz and Bardow's recent LCA of the TSA DAC manufactured by Climeworks finds a CCE from 85.4% to 93.1% for the current demonstration project (with CO_2 capture capacity: 4 kt yr^{-1}) and 88% to 95% for a scale up (with CO_2 capture capacity: 1,000 kt yr^{-1}) supplied by wind energy[20]. Deutz and Bardow quantify the climate benefit of a continuous transition to low-carbon energy but do not evaluate changes in some critical technical parameters, including fan efficiency and DAC plant lifetime[20]. Additional review is presented in Supplementary Note 3.

Goal and scope of the comparative assessment

A comparative and transparent LCA of demonstration projects using HT-Aq and TSA DAC technologies, an extensive sensitivity analysis of crucial improvement options and a detailed estimation of the resource requirements of large-scale DAC deployment compared with other climate change mitigation options have not been published yet. This study fills these gaps by reporting a fully transparent LCA to quantify the comparative life-cycle impacts of two major DAC technologies currently under development: HT-Aq and TSA. These two technologies are currently the only ones that are mature enough and reasonably well documented to be assessed[10,21]. There is still much uncertainty regarding the process-parameter improvement potential, especially for the sorbent regeneration rates, which is crucial to reducing material consumption. Moreover, a prospective assessment is needed to identify the technical conditions under which DAC can become a viable option for CO_2 removal at a climate-relevant scale. The starting point of our analysis is a set of modelled hypothetical new DAC product systems with different electricity and heat supply options other than what is used by the current prototypes. In particular, our LCA assumptions for the HT-Aq technology case include an indirect thermal calciner that releases combustion CO_2 but retains process CO_2. We conduct an attributional comparative LCA of the modelled product systems, and extend the LCA into a sensitivity analysis covering a broad array of relevant improvements in technology, infrastructure (design consideration) and carbon intensity of energy supply. We identify process parameters and energy supply scenarios under which the different DAC technologies are carbon negative from a systems perspective, including a simplified description of CO_2 transport and storage. We also quantify particulate matter (PM_{10}—inhalable particles with diameters that are generally ≤10 μm) emissions, metal depletion, fossil fuel depletion, water depletion and land occupation. We model a scale-up of the two technologies to a capture capacity of 1 Mt yr^{-1} and then linearly on to 1 Gt of CO_2 captured per year to quantify the impacts and resource requirements of a potential large-scale DAC deployment during the coming decades.

Comparative LCA of the modelled DAC product systems

The contributions of the different life-cycle stages to environmental midpoints show a pattern that is typical of energy conversion assets. The relevant stages include the construction phase, operational energy use (heat and electricity) and operational material (sorbent) production, but no single life-cycle stage dominates all the different impact categories. Life-cycle impacts of the HT-Aq DAC demonstration project (reference case) are higher than for the TSA DAC demonstration project (reference case) for the six central categories studied (Fig. 1). To capture and store 1 t of atmospheric CO_2 via HT-Aq and TSA DACCS, 0.58 t (+0.2/−0.03 t) and 0.30 t (+0.02/−0.009 t) CO_2-equivalent (CO_2e) are released to the atmosphere by other parts of the product system, which partially offsets the captured CO_2. Process heat demand to remove the CO_2 from the sorbent is the main driver of life-cycle greenhouse gas emissions (GHG; expressed in CO_2e), and due to the use of natural gas

Table 1 | Technical parameters for the HT-Aq DAC plant for the reference case, best-case, worst-case and scale-up scenarios

Technical and design parameters for HT-Aq	Reference case	Best case	Worst case	Scale up
CO_2 capture rate (%) (share of CO_2 absorbed from air passing through)	42	47	25	70
Single plant CO_2 capture capacity per year (t)	345	442	207	1,000
Plant lifetime (yr)	20	22	15	20
Dimensions of the air contactor (length × breadth × height) (m)	20 × 8 × 200 (scaled down)	20 × 8 × 200 (scaled down)	20 × 8 × 200 (scaled down)	20 × 20 × 200
Number of air contactors	1	1	1	10
Cross section of air inlet (m^2)	30	30	30	4,000
Air inlet speed ($m\,s^{-1}$)	1.4	1.6	1.4	1.8
Efficiency of the fan (%)	68	90	50	56
Sorbent recovery rate (%)	99.90	99.99	99.00	99.90
Mass transfer coefficient (K_L) ($m\,s^{-1}$)	1.50×10^{-3}	2.00×10^{-3}	8.00×10^{-4}	1.50×10^{-3}
KOH required to capture 1 t CO_2 (kg)	2.8	0.25	38	2.8
CaOH required to capture 1 t CO_2 (t)	1.90×10^{-3}	1.93×10^{-4}	2.6×10^{-1}	1.90×10^{-4}
Dimensions of the pellet reactor (diameter, height) (m)	1.2, 12	1.2, 12	1.2, 12	6, 230
Dimensions of the calciner (diameter, height) (m)	0.17, 5	0.17, 5	0.17, 5	6, 165
Dimensions of the slaker (diameter, height) (m)	0.15, 5	0.15, 5	0.15, 5	6, 20.7
Total electrical energy required ($kWh\,t^{-1}$ CO_2 captured)	345	337	449	534
Total heat required ($GJ\,t^{-1}$ CO_2 captured)	4.47	4.05	4.47	4.05
Transportation distance from site of production to site of installation to end-of-life treatment facility (km)	50	50	50	50
Energy required to compress 1 t CO_2 ($kWh\,t^{-1}$ CO_2 captured)	111	111	111	111
Pipeline required to transport CO_2 (km)	300	300	300	300
Energy required to inject 1 t CO_2 in geological storage ($kWh\,t^{-1}$ CO_2 captured)	7	7	7	7

Best-case and worst-case results are shown in Fig. 1. See Supplementary Data 1, sheet 2 for details.

in DAC units, it is also the main contributor to fossil fuel depletion. The production of iron and steel alloying elements used for DAC unit manufacturing dominate metal depletion. The construction and eventual demolition of DAC units represent the largest contribution to PM_{10} emissions. Water depletion is driven mainly by the construction and demolition of HT-Aq DAC units and electricity demand during the use phase of TSA DAC. The indirect (supply chain) land occupation of HT-Aq DAC is approximately four times higher than that of TSA DAC, mainly due to the production and end-of-life phase of DAC units and adsorbent.

Even with unfavourable technological parameters (sensitivity analysis worst case; see Tables 1 and 2 for parameters), TSA and HT-Aq DACCS remain net carbon negative with worst-case CCE of 30% and 15%, respectively. Capture phase energy, DAC lifetime and the carbon capture capacity and efficient regeneration of the sorbent noticeably affect DAC performance and life-cycle impacts, as shown by the wide range between the worst and the best sensitivity cases (Fig. 1).

Important parameters for climate performance

A detailed assessment of the main parameter changes shows a number of improvement options but also trade-offs between impact categories (Fig. 2). (More sensitivity cases are shown in Supplementary Fig. 7.)

High sorbent recovery rate

High HT-Aq sorbent recovery rates, which govern sorbent lifetime, are required for curbing environmental impacts. A reduction of the sorbent recovery rate from its base value 99.90% (1,000 capture cycles) to 99.00% (100 cycles) increases the impact on climate change by 15% and PM_{10} formation by 25% while an improvement to 99.99% (99.984% is reported by the US National Academies[22] and Socolow et al.[3]) decreases the impact on climate change by only 2–3%. Increasing the lifetime of the polyethylenimide (PEI)-infused silica sorbent for TSA from 0.5 to 3 years lowers the life-cycle climate impact by approximately 15% and PM_{10} emissions by 38%.

Low-carbon electricity and heat supply

A low-carbon energy supply (here: 70 g CO_2e kWh^{-1}) can make the TSA DAC demonstration plant with storage almost 86% carbon efficient and HT-Aq DAC almost 73% carbon efficient (CCE values), corresponding to life-cycle GHG emissions per ton of CO_2 captured of 0.15 t for TSA DAC and 0.26 t for HT-Aq DAC. Metal depletion increases by 10% and 70% for HT-Aq DACCS and TSA DACCS reference cases, respectively, due to the higher specific metal demand of low-carbon renewable electricity[23]. Adding high-temperature heat pumps to the TSA DACCS reference case increases its CCE to 80%, and heat pumps combined with LCE supply yield a CCE of 90%. However, additional resource extraction and material production for building the low-carbon heat and electricity supply infrastructure leads to an increase of PM_{10} formation, metal depletion, water depletion and land occupation. Emissions in the aforementioned categories noticeably increase by 25–250% when high-temperature heat pumps are used for TSA DACCS along with LCE supply.

For both DAC technologies, LCE supply for sorbent production only (and not the CO_2 capture phase) results in life-cycle impact reductions from 1% to 20% in all impact categories except land occupation for HT-Aq DAC (+50%).

Scaling up DAC to the megaton and gigaton levels

To be relevant at the global scale, annual DAC capture flows in the gigaton range are needed by 2050 and beyond. Here, impact results for a process-based scale up to 1 Mt yr^{-1} are reported for HT-Aq DAC (see Supplementary Data 1, tables B.3 and G.1 for calculations).

Table 2 | Technical parameters for the TSA DAC plant for the reference case, best-case, worst-case and scale-up scenarios

Technical and design parameters for TSA	Reference case	Best case	Worst case	Scale up
CO_2 capture rate (%)	90	100	70	90
Single unit CO_2 capture capacity per year (t)	50	50	50	1,000
Number of air contactors	1	1	1	20,000
Lifetime of the air contactor (yr)	20	22	15	20
Carbon capture capacity of the sorbent (%)	10	12	4	10
Lifetime of sorbent (yr)	1	3	0.5	1
Sorbent required per t CO_2 captured (kg)	6.9	1.5	34.2	6.9
Air velocity ($m\,s^{-1}$)	4	3	5	4
Fan efficiency (%)	68	90	40	68
Total electrical energy required (kWh)	180	130	350	180
Adsorption temperature (°C)	25	20	27	25
Desorption temperature (°C)	100	80	130	100
Heat recovery rate (%)	75	90	10	75
Heat required (GJ t^{-1} CO_2 captured)	2.6	2.3	6.2	2.6
Transportation from site of production to site of installation to end-of-life treatment facility (km)	50	50	50	50
Energy required to compress 1 t CO_2 (kWh t^{-1} CO_2)	111	111	111	111
Pipeline required to transport CO_2 (km)	300	300	300	300
Energy required to inject 1 t CO_2 in geological storage (kWh t^{-1} CO_2 captured)	7	7	7	7

Best-case and worst-case results are shown in Fig. 1. See Supplementary Data 2, sheet 2 for details.

These 1 Mt units can then be deployed in large numbers to reach capture capacities in the Gt range.

For TSA DAC, the modular nature of the capture devices allows for a linear scale-up of small, 40-ft shipping container-sized (12.19 m long × 2.44 m wide × 2.59 m high) units with a stack height of six and only minimal contributions from concrete foundations. Hence, impacts per ton CO_2 captured do not change (Fig. 3).

As a result of efficiency of scale, material intensity for infrastructure requirement decreases per ton of CO_2 captured in the case of the HT-Aq DAC scale-up scenario, leading to an overall decrease in the environmental impact per ton CO_2 captured (Fig. 3). The impacts in the categories of climate change, PM_{10} formation, metal depletion, fossil depletion, water depletion and land occupation decrease by 22%, 47%, 95%, 18%, 46% and 50%, respectively. Another major reduction of HT-Aq DAC (scale-up) climate impacts can be achieved with LCE supply (−73%) but at the expense of higher water depletion (+80%) and land occupation (+77%).

Utilizing either a high-temperature heat pump or LCE for TSA DACCS reduces the non-capture life-cycle climate impact by 80% and 86%. As efficiency technology, the heat pump outperforms the LCE scenario by a factor of six for land occupation and for water depletion. The heat pump and LCE combined scenario for TSA DAC has the highest CCE (90%) of all scenarios and sensitivity cases studied. Its land use compared with the baseline increases moderately, by a factor of four.

DAC carbon capture efficiency

For technology comparison and scenario modelling purposes, we model several hypothetical new DAC product systems with different electricity and heat supply options other than what is used by the current prototypes. For the modelled DAC product systems, co-capture of CO_2 from natural gas combustion is excluded and hence, the heat required for desorption of CO_2 and regeneration of the sorbent material is the single largest contributor to both climate change and fossil fuel depletion. It is also a major cost factor[11,15,24]. The results emphasize the importance of utilizing LCE sources during the use phase of both DAC technologies, for heat as well as for electricity to operate the fans and pumps, which can result in CCE values as high as 90% (for TSA DAC). Using low-carbon renewable energy can bring down costs, especially since the timing of the different DAC process steps can be shifted to times of LCE abundance to some extent, though additional battery storage will be needed[10]. Our analysis shows that next to using LCE, a longer DAC asset lifetime, a higher sorbent recovery rate per lifetime, the installation of high-temperature heat pumps for TSA DAC and the use of operational chemicals (KOH, calcium hydroxide ($Ca(OH)_2$) and PEI) with low-carbon footprint, are major technology learning goals, and supply chain transformations are important for reducing negative DAC impacts. At the smaller scale, medium-temperature (approximately 100–120 °C) and low-carbon heat from waste to energy can be used to decrease the environmental impact of TSA DAC[24]. High-temperature solar heat and its storage is a proven technology. Concentrated solar power generation is projected to increase from 15 TWh in 2019 to 180 TWh in 2050[25], and this rapid scale-up could make low-carbon, high-temperature heat available for HT-Aq DAC as well[26] at a reasonable cost. A comprehensive assessment of this option, however, is beyond the scope of this paper.

Even though the energy required per ton of CO_2 captured is comparable for both DAC technologies, TSA DAC has a lower environmental footprint than HT-Aq DAC across the impact categories studied. The main reason is that unlike TSA DAC, HT-Aq DAC requires high-temperature heat and continuous processing of very large mass flow, which translates into relatively high energy demand and infrastructure requirements. Originally, HT-Aq DAC was designed so that readily available commercial-scale reactors could be used[27]. Given the low concentration of CO_2 in the atmosphere, the high throughput of HT-Aq DAC and the medium-temperature DAC technologies available now, HT-Aq DAC does not appear as a strong competitor for a potential large-scale roll-out of DAC from a life-cycle perspective.

Sorbent lifetime and consumption

The operational material demand and waste generation of DAC is substantial, and the sorbent recovery rate (HT-Aq) and lifetime

(TSA) have a major impact on the DAC life-cycle environmental footprints[11,12]. With the sensitivity analysis cases of HT-Aq DAC sorbent recovery rates of 99.00%, 99.90% and 99.99% (Table 1) and TSA DAC sorbent lifetimes of 0.5, 1 and 3 yr (Supplementary Data 2, table F.4b), the resulting average KOH consumption is 38 kg, 2.8 kg and 0.25 kg for HT-Aq DAC and 14 kg, 7 kg and 2.3 kg of PEI for TSA DAC per ton of CO_2 captured. For a 1 Gt yr^{-1} TSA DAC scale-up, additional demand for ammonia for the sorbent production would be between 1.8 and 2.7 Mt yr^{-1} (1–1.5% of current global production), and for HT-Aq DAC, additional KOH demand would be 2.6–2.8 Mt yr^{-1} or 100–108% of current global production (0.28–1.8 Mt yr^{-1} or 0.4–2% of current global production if NaOH is chosen instead) under optimistic and current conditions (see Supplementary Data 3, sheets S1 and S2). These numbers are considerably lower than previous estimates published[16,17]. They correspond well with the data reported by Deutz and Bardow[20] for TSA and with the range reported for HT-Aq (Supplementary Data 3, sheet S1). The problem of high material consumption and related waste-generation impacts for large-scale DAC remains but at a much smaller scale than feared[17].

Cross-strategy comparison for 1 Gt CO_2 captured or avoided

To design technology development roadmaps for a low-carbon society, it is crucial to understand the energy requirements and adverse environmental impacts of DAC compared with other GHG mitigation options based on low-carbon renewable energy. We therefore compare energy and resource demand of the scale-up scenarios for HT-Aq DAC and TSA DAC to the additional energy required for the implementation of carbon capture and storage (CCS) technologies at coal and natural gas power plants[28], BECCS[29], a shift from gasoline and diesel vehicles to electric vehicles[30–32] and a shift from natural gas boilers to electric boilers and heat pumps[30–32].

This comparison is not an accurate comparative LCA of all the different pathways but a first estimate of the general trend. For this simple comparison, the technologies were assumed to scale linearly, assuming no technology learning. They are either modelled as scale-ups of small units at the megaton level (like the ones modelled for DAC in our paper) or scaled down from even larger-scale estimates, like for BECCS. The results are reported in Fig. 4, where life-cycle net GHG reduction or removal (*y* axis) is shown as a function of the additional low-carbon or fossil energy deployed for the different pathways (*x* axis) (Details in Supplementary Note 5, where a case with technology learning is illustrated also).

TSA DAC has a renewable energy demand similar to the electrification of the vehicle fleet for the same GHG reduction or removal (Fig. 4, see Supplementary Data 3, sheets S3 and S4 for details), with only 97% of the land use but much higher material demand (factor 5) (Table 3). The renewable energy input for TSA DAC is not too different from the energy needed for building sector decarbonization via heat pumps (Fig. 4). The annual PM_{10} emissions for large-scale (1 Gt CO_2 captured per year) roll-out of TSA DAC are around 180 kt (Table 3), which is 30% of the annual PM_{10} emissions of the global vehicle fleet[33].

For the same GHG mitigation, HT-Aq DAC, however, requires about 65% more renewable energy than a switch from gasoline to electric vehicles and 20% more than a shift from natural gas boilers to heat pumps. Hence, it would be a comparatively inefficient carbon sequestration investment as long as the transport and building sectors are not yet decarbonized. BECCS is different in that it provides final energy (for example, electricity) as a co-product, which explains part of the high energy demand.

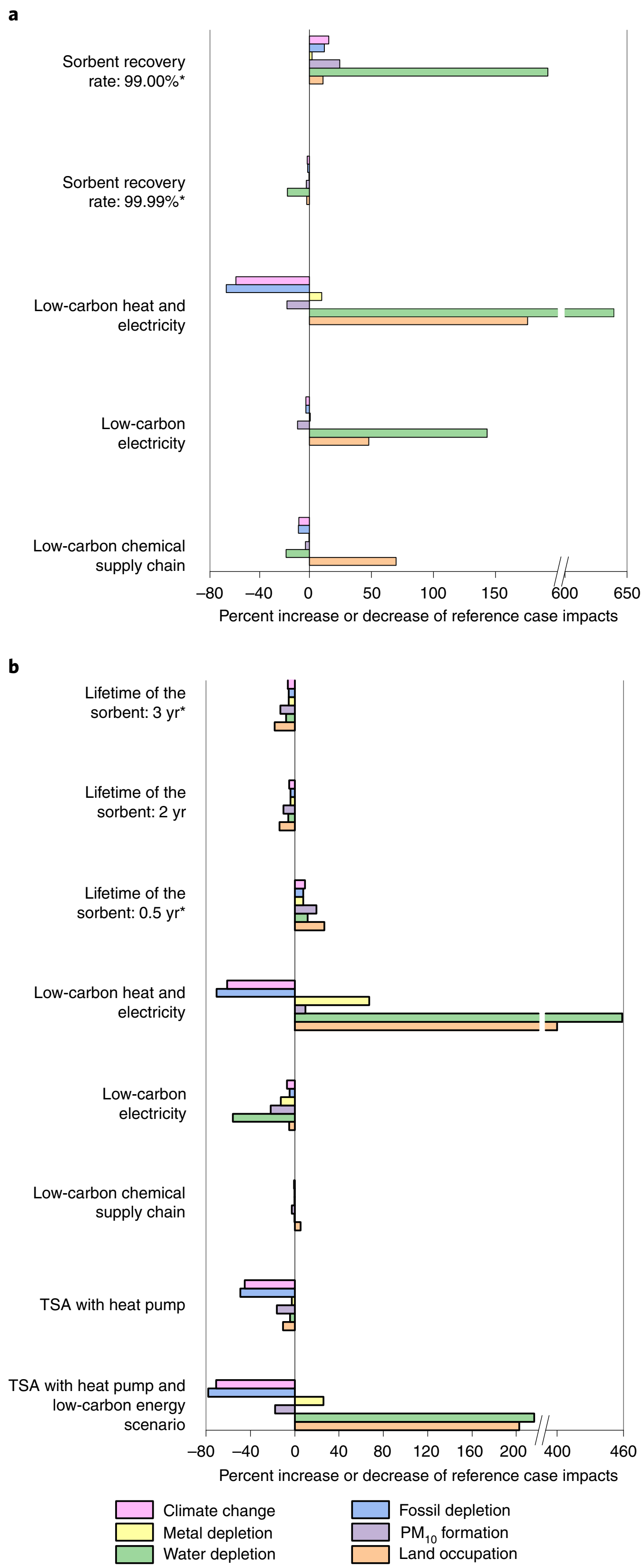

Fig. 2 | Sensitivity analysis of main environmental pressure indicators as a consequence of plausible changes in crucial process parameters. a,b, The percent increase (right) or decrease (left) of the reference case impacts for HT-Aq DAC (**a**) and TSA DAC (**b**) for 1 t atmospheric CO_2 captured. The *x*-axis break is from 200% to 600% (**a**) and 220% to 390% (**b**). Parameter changes indicated with an asterisk are included in the best case-worst case sensitivity analysis in Fig. 1; the other parameter changes represent further technology and energy supply improvement.

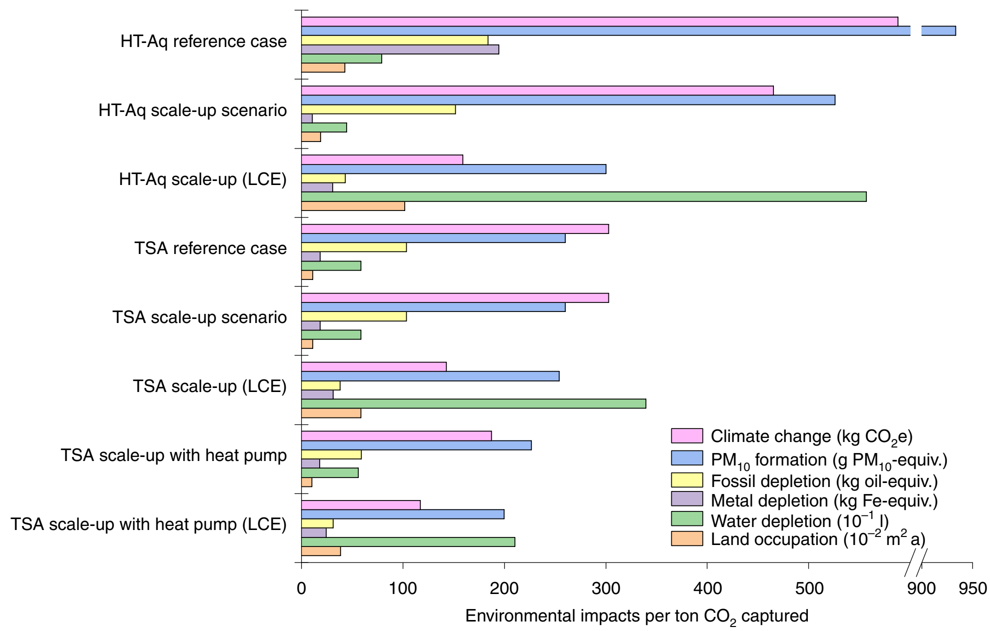

Fig. 3 | Scale-up scenario for 1 Mt CO_2 captured per year for the two DAC technologies. LCE and the use of a high-temperature heat pump (TSA only) are included. Results are shown for the impact categories climate change, metal and fossil depletion, PM_{10} emission, water depletion and land occupation. The *x*-axis break is from 600 to 900.

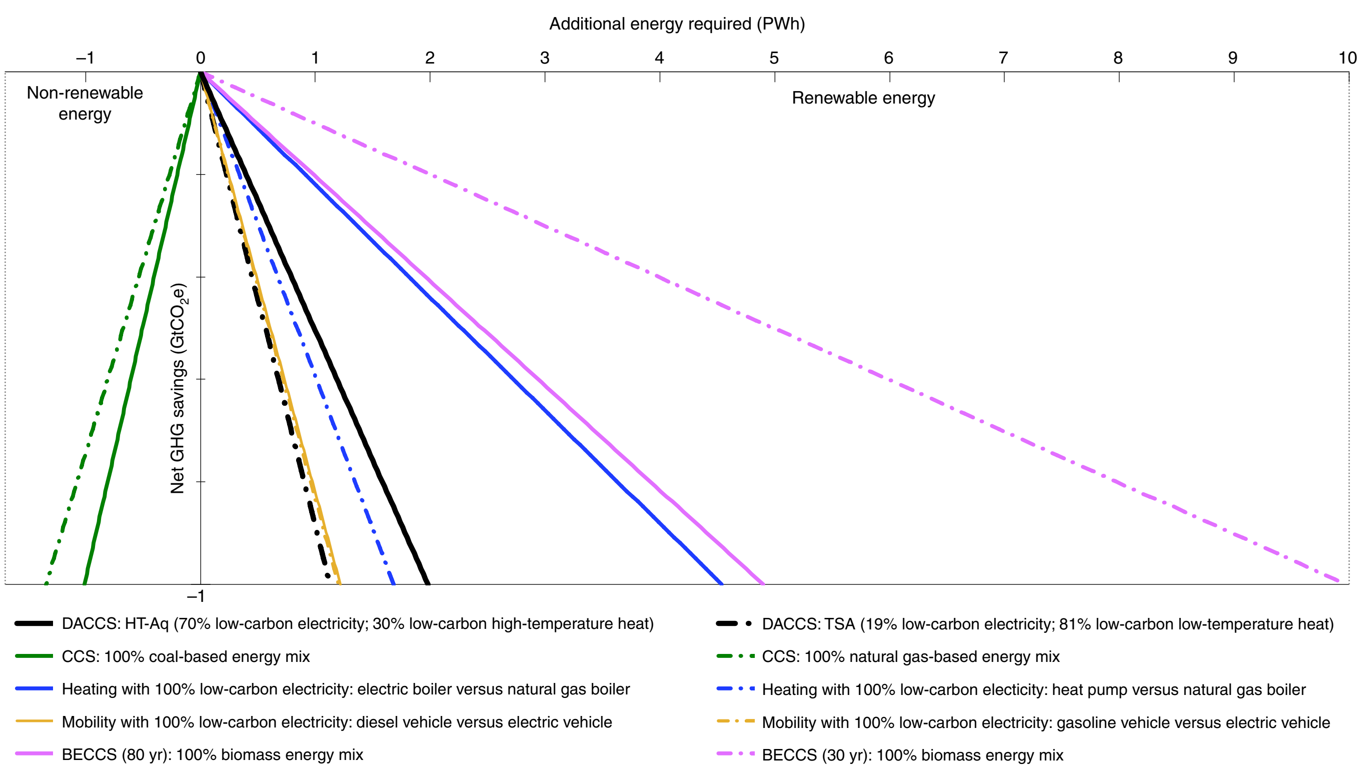

Fig. 4 | Technology scale-up comparison. The figure shows net GHG reductions achieved by additional energy use for DAC as well as for other carbon-removal technologies, fuel shift applications and end-of-pipe solutions. Net emissions are reported as GHG savings (Gt CO_2e) from a life-cycle perspective.

Figure 4 illustrates the energy flows that are partly primary (CCS, BECCS) and partly final (DAC, mobility, heating) because it shows the energy stage that determines environmental impacts and resource use for each technology. For DAC, electricity and heat demand were added together. A version of Fig. 4 with all energy flows converted to exergy is shown in Supplementary Fig. 11. It shows that due to the large share of heat in the operational energy requirement, TSA DAC exergy input is only about 30% of the exergy demand from renewable sources of a switch from gasoline to electric vehicles.

Table 3 | Corresponding impacts on material (steel, concrete, copper and aluminium), land and water use and PM_{10} emissions, including life-cycle impacts of LCE supply

For net removal of 1 Gt CO_2e of GHG	DAC: HT-Aq (removal)	DAC: TSA (removal)	BECCS (80 yr, removal)	Shift from natural gas boiler to heat pump (mitigation)	Shift from gasoline to electric vehicles (mitigation)
Additional renewable energy demand (PWh)	2	1	5	1.6	1
Material use (Mt)	17	36	8	11	8
Land occupation ($10^3\,km^2a$)	20	11.5	1,100	16	12
PM_{10} (kt PM_{10} equivalent)	170	180	380	84	60
Water use (km^3)	5	3	0 (rainfed)	4	3
Sorbent required (Mt)	3	7			

For calculation details, see Supplementary Data 3, sheet S3.

Additional material demand (steel, concrete, copper and aluminium) for 1 Gt CO_2 sequestered per year by TSA DAC and HT-Aq DAC is 4–6 Mt, 12–29 Mt, 0.3–0.4 Mt and 0.6 Mt, respectively, which is well below 1% (steel, concrete) or between 1% and 2% (copper, aluminium) of the current global annual production of these materials (see Supplementary Data 3, Fig. 4 and Deutz and Bardow[20]). BECCS comes with very large land requirements (Table 3) that are likely to compromise biodiversity and other land-use sustainability targets if deployed at large scale[34,35], especially if sourced from land that does not require irrigation[34].

Consequences for the future roll-out of DACCS

As DAC opens up a new technological pathway and does not provide any service other than CO_2 sequestration, the development pattern of DAC would differ greatly from technology-based emissions mitigation. DAC technologies would require additional land and investment.

The capital costs of DAC are substantial[10,11,15], and the operational costs are dominated by energy costs and heat requirements, in particular[10]. Cheap low-carbon heat sources, such as waste heat, and the implementation of a highly efficient heat recovery system can lower capture costs[10]. A 1 Gt yr^{-1} roll-out of the DAC units would require (on site) a minimum of 85 km^2 of land for TSA DAC and 400 km^2 for HT-Aq DAC, plus 20,000 km^2 and 11,500 km^2 for supplying the required electricity from photovoltaics and the DAC supply chains (Supplementary Data 3, sheet S3). The DAC units do not require any specific land type; they can be deployed on degraded or barren land provided the local air quality and the CO_2 transport costs are not prohibitive. The main contribution to the DAC land footprint, however, stems from the renewable energy supply (Fig. 4 and Table 3).

Heat generation from natural gas combustion needs to be replaced by low-carbon heat sources before DAC can be deployed at large scale. As low-cost LCE will continue to be a scarce resource, efficient service-providing mitigation strategies that avoid (additional) energy demand have a clear advantage over mere removal technologies (Fig. 4). Since their infrastructure material, land-use and PM_{10} impacts are similar or smaller than those of the removal technologies, they remain key to deep decarbonization as long as their potential is not fully seized. As renewable energy becomes more abundant, the business case for efficient carbon removal may improve, especially if DAC is used to compensate for difficult-to-mitigate emissions in the chemical[31], steel or cement industries[36,37] or other residual emissions such as those from agriculture. Detailed life-cycle and cost assessments, combined with multicriteria optimization involving desirability constraints on land, water and material use, can identify optimal transformation pathways and determine the share of DAC in a portfolio of climate change mitigation strategies of nations or organizations.

Limitations and outlook

Due to the limited data availability and the early development stage of DAC technology, a number of informed estimates had to be made for both the scale-up scenario and the lifetime of the sorbent, in particular. The results directly depend on these estimates and may differ from the actual life-cycle impacts of the current DAC demonstration plants. Future assessments of the environmental performance of DAC can be improved by developing calibrated accurate process models using chemical engineering tools and updating the process database as the technology matures and more information becomes available[20]. Further research is necessary to systematically compare the many different energy use and decarbonization pathways leading to net negative emissions, including carbon capture and utilization[29]. Continued assessments of the costs[10,15] as well as the political, economic and ethical implications[38] of a potential large-scale DAC deployment are highly relevant, and the results of our work can be used to inform such assessments. The costs, as well as material and energy repercussions of mitigating salient environmental impacts, such as PM_{10} and water depletion, need to be estimated as well. The scale-up scenario hinges upon the availability of large-scale permanent geological CO_2 storage, the capacity and permanence of which is still under investigation[39]. Next to the land use for the supply of low-carbon renewable energy to operate DAC, the land use of the DAC construction and material supply chain is substantial and needs to be considered in assessments of large-scale DAC deployment. Under the conditions identified above, DAC can be an effective negative emissions technology with much lower impacts on land and water use than BECCS, in particular, and energy requirements comparable to those of electric vehicles and heat pumps. Requirements on chemical input are massive but pose no substantial constraint even if DAC technologies sequester more than 1 Gt CO_2 per year. Still, its large-scale deployment adds to global land, water and resource use to a considerable extent while providing no service other than removing CO_2 from the atmosphere. As long as low-carbon renewable energy is a globally scarce resource and emissions mitigation options are abundant, mitigation can be expected to be more cost effective and create larger environmental co-benefits than a DAC roll-out.

Methods

Technical details for LCA. LCA is a tool widely used to evaluate the environmental impact of individual services from a systems perspective. It is a scientific standard to evaluate both mature and emerging technologies and therefore was chosen for assessing DAC technologies[40]. To conduct the LCA in accordance with both the scientific and International Organization for Standardization (ISO) standards 14040 and 14044[41,42], the OpenLCA software version 1.8 was used together with Ecoinvent database version 3.5. The life-cycle impact assessment method used for the study is ReCiPe Midpoint (H) w/o LT, which is a standard in scientific LCA research that indicates how different environmental mechanisms are affected by various anthropogenic emissions over a 100 year time horizon for most mechanisms and without considering uncertain long-term effects. Below, the modelling choices and data compilation for the DACCS case studies and scenarios are reported, following the steps given by the ISO standard.

Goal and scope of the LCA study. The goal of the LCA study is twofold. First, to conduct an attributional cradle-to-cradle (from sourcing of natural resources to the recycling and disposal and recycling of the assets at the end of their useful life) LCA of the current HT-Aq and TSA DAC pilot plants, with a CO_2 sequestration capacity of 345 t yr^{-1} (calculated, Supplementary Data 1, table G.1) and 50 t yr^{-1} (ref. [43]), respectively. This work includes an uncertainty analysis in the form of a Monte Carlo simulation for the different process parameters and the Ecoinvent background database, as well as a sensitivity analysis for different technology choices, improvement options (Tables 1 and 2) and energy supply scenarios. The second goal is to conduct a prospective LCA for a scale-up and technology learning scenario of the aforementioned DAC technologies to a CO_2 capture capacity of 1 Mt yr^{-1}, followed by a linear scale-up to 1 Gt yr^{-1} to estimate the resource requirements of a large-scale DAC deployment.

Scope and system boundary. The system boundary includes the material and energy input required for the production, operation and end-of-life treatment of the DAC air contactor, sorbent material and other auxiliary infrastructure required to capture CO_2 from ambient air and store it in a geological sink. Transportation of materials, components and waste are also included. The exchange between the product system and the natural environment includes resource use and emissions to the environment relevant for the impact categories listed below. The functional units for this study are 1 ton of CO_2 captured and stored for the attributional LCA and sensitivity analysis and 1 Mt and 1 Gt of CO_2 captured and stored per year for the upscaling.

Selection of impact categories. In addition to the focus on global warming, von der Assen et al.[44] recommended quantifying fossil fuel depletion for the environmental assessment of carbon capture and utilization technologies[45] to quantify the extent of decoupling from these energy sources. Other relevant categories for LCA studies of new energy technologies include metal depletion and PM_{10} emissions[45], which are also central to the debate about the sustainability contribution of DAC[17]. Following this guidance, we report here life-cycle impacts of the DAC technologies for the following environmental midpoints: climate change, fossil depletion, metal depletion, PM_{10} emissions, water depletion and land occupation. Results for the other ReCiPe midpoint impact categories are available in Supplementary Data 1, sheet F for HT-Aq DAC and Supplementary Data 2, sheet E for TSA DAC.

Life-cycle CCE. The most salient DAC-specific life-cycle indicator is life-cycle CCE (equation (1)), which is defined as minus the net (life-cycle) CO_2 balance of the technology divided by the amount of CO_2 captured from ambient air. Its maximum value is 100%, which means that no GHG emissions occur elsewhere in the system due to the capture. A negative CCE means that the life-cycle emissions of building and operating DAC are larger than the CO_2 captured (equation (1)). Net (life-cycle) CO_2 emissions are defined as the sum of manufacturing (GHG_{Manuf}), direct and indirect operational emissions ($GHG_{Operation}$) and end-of-life impacts (GHG_{EoL}) minus the CO_2 captured.

$$\mathrm{CCE}\,(\%) = 100 \times \frac{-\text{Life-cycle GHG}}{\mathrm{CO_2}\ \text{captured}} \tag{1}$$

$$\begin{aligned}\text{Life-cycle GHG} &= \mathrm{GHG}_{\mathrm{Manuf}} + \mathrm{GHG}_{\mathrm{operation}} \\ &\quad + \mathrm{GHG}_{\mathrm{EoL}} - \mathrm{CO_2}\ \text{captured}\end{aligned}$$

System and technology description summary. A general overview of the examined technologies and their central performance parameters is given here. Full details on data and calculations are given in Supplementary Data 1 (for HT-Aq DAC) and Supplementary Data 2 (for TSA DAC).

HT-Aq DAC. The product system for the HT-Aq DAC uses information published by Carbon Engineering and adds several distinct changes to Carbon Engineering's DAC technology, resulting in a hypothetical new product system without co-capture of fuel-related CO_2 in the calcination step[15,46] (see Supplementary Note 2). This modelling decision allows us to compare the product systems of HT-Aq and TSA DAC, however, the baseline results of HT-Aq DAC are not representative for Carbon Engineering's DAC technology. The adsorption and desorption of CO_2 takes place at ambient temperature and 900 °C, respectively[10]. The DAC unit consists of four major infrastructural parts: air contactor, pellet reactor, calciner and slaker, which are used for the capture of CO_2 and regeneration of the sorbent (1 M KOH)[14]. The KOH solution flows over a 3-m thick polyvinylchloride (PVC) mesh in the air contactor. The product system also encompasses the material and energy required for production and end-of-life treatment of a CO_2 compressor, pipelines to transfer the compressed CO_2 and injection into a geological storage (Table 1). A sensitivity analysis was conducted for different DAC asset lifetimes, different sorbent recovery rates, low-carbon energy mixes and a low-carbon supply chain for chemicals (Table 1).

The exact sorbent recovery rate per lifetime is not known, but several boundaries and estimates exist in the literature (Supplementary Note 2)[3,22]. For the reference case, it was assumed that 99.90% of the sorbent and recovery chemicals, KOH and $Ca(OH)_2$, that are circulating in the DAC plant are recovered at the end of the capture cycle. The sensitivity analysis includes a change of the sorbent recovery rate up to 99.99% and down to 99.00%.

TSA DAC. The product system for TSA DAC is based on the commercialized DAC units manufactured by the Swiss company Climeworks[43]. The analysed technology has a capture capacity of 50 t CO_2 per year, a carbon capture rate of 90% and an asset lifetime of 20 years. The adsorption and desorption of CO_2 take place at 25 °C and 100 °C, respectively. The main components modelled for the capture of CO_2 and regeneration of the sorbent material are the air contactor, the sorbent material and the heat exchange unit. Approximately 340 kg of sorbent (assumed to be PEI because of its high resistance to degradation from contact with water or air[11] and modelled as monoethanolamine as closest proxy in the Ecoinvent background database; details in Supplementary Note 4) are needed every year, corresponding to an average sorbent loss of 7 kg per ton of CO_2 captured. The sorbent regeneration is energy intensive, and for the reference case, it is assumed that the plant is equipped with a heat recovery system working at an efficiency of 75%. The technical parameters for TSA DAC are given in Table 2.

A sensitivity analysis was conducted for the lifetime of TSA DAC, the PEI-silica sorbent consumption, a low-carbon energy mix, a low-carbon supply chain for the sorbent and utilization of heat pump to provide low-temperature heat.

Lifetime of the PEI-silica sorbent. For the reference case, it was assumed that the PEI-silica sorbent has a lifetime of 1 year, which means that on average, 7 kg of sorbent are required to capture 1 t CO_2. A sensitivity analysis was conducted for sorbent lifetimes of 6 months, 2 years and 3 years, under the reference case conditions, corresponding to a sorbent consumption of 14 kg, 3 kg and 2 kg per 1 t CO_2 captured (see Supplementary Data 2, table F.4).

Compression, transfer and storage of captured CO_2. For both capture technologies, the captured CO_2 is compressed on site by a compressor that requires 111 kWh of electricity per 1 t CO_2 (refs. [18,47]). It is further assumed that the compressed CO_2 is transported by pipeline over a distance of 300 km and then injected into geological wells, requiring an injection energy of 7 kWh t^{-1} CO_2 (ref. [47]).

Accounting for the uncertainty of the reference case. A Monte Carlo analysis was used to propagate the uncertainty of the process descriptions in the background database (Ecoinvent) and the foreground data on, for example, steel, aluminium, cement and electronics requirements of the DACs into the environmental pressure indicators. The foreground data have uncertainty ranges of 10–30% (see Supplementary Data 1, table K.1 and Supplementary Data 2, table G.1), and the uncertainty of the background database is documented in the Ecoinvent reports[48]. The Monte Carlo analysis was conducted using the Monte Carlo feature of OpenLCA with 1,000 individual runs for each technology, and the 5th and 95th percentiles are reported here.

Process-parameter uncertainty and technology learning. The total CO_2 capture capacity of a DAC asset mainly depends on the lifetime of the DAC units, sorbent lifetime, carbon capture rate and specific heat and electricity requirements. Two sensitivity cases are developed for each DAC technology by setting salient technical parameters to the highest and lowest values found in the literature, where possible, or to plausible alternatives, where necessary. Depending on whether the carbon capture rate increases or decreases compared with the reference case, the parameter choice was named best technology case (best case) or worst technology case (worst case). The ancillary calculations for the different cases are documented in Supplementary Data 1 (tables G.1, I.1 and J.1) and Supplementary Data 2 (tables F.1 and F.3).

Common sensitivity analysis for HT-Aq DAC and TSA DAC. The sensitivity analysis includes a LCE mix (at 70 g CO_2e kWh^{-1}) and a low-carbon supply chain for the sorbents, that is, KOH and $Ca(OH)_2$ for HT-Aq DAC and the PEI-silica for TSA DAC. Site-specific electricity mixes used in the reference case are replaced with the alternative low-carbon electricity mix. The heat source for the reference case for the DAC technologies is natural gas, whereas the heat source in the LCE case is solar-thermal energy. For the reference cases of the DAC technologies, sorbent production was modelled after Ecoinvent processes, representing a global average. For the sensitivity analysis, the sorbent production process for TSA is changed from the global average to a Switzerland-specific supply chain, which uses a comparatively less carbon-intensive energy supply for production. Similarly, for HT-Aq DAC, the sorbent production representing a global average is replaced by a Canada-specific supply chain, which use a comparatively low-carbon-intensive energy supply for production. The decision to use country-specific production processes is based on the current location of the pilot plants for these technologies.

Use of heat pump. For TSA DAC, it is possible to reduce the heat demand by utilizing high-temperature heat pumps with a coefficient of performance 2.5 or even higher[20]. The heat required for the TSA scenario with heat pump is calculated as 1.04 GJ (calculation in Supplementary Data 2, table B.7). Heat pumps are included in the sensitivity and upscaling analyses (Figs. 2 and 3), both with default energy (reference case) and with LCE supply.

Scenario analysis with scale up. *HT-Aq DAC.* A scale-up scenario was drafted that represents the infrastructure and energy requirements of an increase of the capacity from 345 t to 1 Mt CO_2 captured per year (Table 1, right column). For 1 Mt yr^{-1}, ten air contactors would be needed, each with the dimensions of 20 m width, 20 m height and 200 m length[15]. The required volumes of the pellet reactor, calciner and slaker are noted in Table 1. The process heat requirements were taken from Keith et al.[15], and electricity requirements were scaled based on air inlet area and fan efficiency. The DAC asset lifetime was kept at 20 years and the chemical recovery at 99.90%. References and calculations are provided in Supplementary Data 1 (table B.3 and G.1). Both standard and LCE supply for the DAC operation are assessed.

TSA DAC. A scale-up scenario was drafted to estimate the infrastructure, energy and material requirements of a transition from 50 to one million t CO_2 captured per year. Since the Climeworks TSA DAC units are modular[43], the 1 Mt scale up simply consists of approximately 20,000 TSA DAC units in standard containers as modelled in the reference case. The asset lifetime stays at 20 years and the sorbent lifetime at 1 year. This scale-up scenario is combined with four sensitivity cases: (1) default electricity mix, (2) low-carbon electricity, (3) high-temperature heat pump with default electricity and (4) high-temperature heat pump with low-carbon electricity.

Data availability

The complete process inventories for the two DAC technologies studied are available and documented in Supplementary Data 1 and 2. The data and assumptions behind the 1 Gt scale-up analysis are documented in Supplementary Data 3, as well as all numerical values plotted in the figures.

Code availability

The OpenLCA process models for different cases of HT-Aq DAC and TSA DAC are provided on Zenodo at https://doi.org/10.5281/zenodo.5500477. To rerun the LCA calculations, OpenLCA (freeware) and the Ecoinvent 3.5 database (license required) need to be installed on a standard desktop computer or laptop with at least 8 GB RAM.

Received: 14 December 2020; Accepted: 17 September 2021;
Published online: 28 October 2021

References

1. IPCC *Climate Change 2014: Synthesis Report* (eds Core Writing Team, Pachauri, R. K. & Meyer, L. A.) (IPCC, 2014).
2. IPCC *Special Report on Global Warming of 1.5 °C* (eds Masson-Delmotte, V. et al.) (WMO, 2018).
3. Socolow, R. et al. *Direct Air Capture of CO_2 With Chemicals: A Technology Assessment for the APS Panel on Public Affairs* (American Physical Society, 2011).
4. Minx, J. C. et al. Negative emissions—Part 1: research landscape and synthesis. *Environ. Res. Lett.* **13**, 63001 (2018).
5. Hanna, R., Abdulla, A., Xu, Y. & Victor, D. G. Emergency deployment of direct air capture as a response to the climate crisis. *Nat. Commun.* **12**, 368 (2021).
6. IPCC *Special Report on Carbon Dioxide Capture and Storage* (eds Metz, B. et al.) (Cambridge Univ. Press, 2005).
7. Wang, T., Lackner, K. S. & Wright, A. Moisture swing sorbent for carbon dioxide capture from ambient air. *Environ. Sci. Technol.* **45**, 6670–6675 (2011).
8. Marcucci, A., Kypreos, S. & Panos, E. The road to achieving the long-term Paris targets: energy transition and the role of direct air capture. *Climatic Change* **144**, 181–193 (2017).
9. Viebahn, P., Scholz, A. & Zelt, O. The potential role of direct air capture in the German energy research program—results of a multi-dimensional analysis. *Energies* **12**, 3443 (2019).
10. Fasihi, M., Efimova, O. & Breyer, C. Techno-economic assessment of CO_2 direct air capture plants. *J. Clean. Prod.* **224**, 957–980 (2019).
11. Sanz-Pérez, E. S., Murdock, C. R., Didas, S. A. & Jones, C. W. Direct capture of CO_2 from ambient air. *Chem. Rev.* **116**, 11840–11876 (2016).
12. Salmón, I., Cambier, N. & Luis, P. CO_2 capture by alkaline solution for carbonate production: a comparison between a packed column and a membrane contactor. *Appl. Sci.* **8**, 996 (2018).
13. House, K. Z. et al. Economic and energetic analysis of capturing CO_2 from ambient air. *Proc. Natl Acad. Sci. USA* **108**, 20428–20433 (2011).
14. Fuhrman, J. et al. Food-energy-water implications of negative emissions technologies in a + 1.5 °C future. *Nat. Clim. Change* **10**, 920–927 (2020).
15. Keith, D. W., Holmes, G., St. Angelo, D. & Heidel, K. A process for capturing CO_2 from the atmosphere. *Joule* **2**, 1573–1594 (2018).
16. Realmonte, G. et al. Reply to 'High energy and materials requirement for direct air capture calls for further analysis and R&D'. *Nat. Commun.* **11**, 3286 (2020).
17. Chatterjee, S. & Huang, K.-W. Unrealistic energy and materials requirement for direct air capture in deep mitigation pathways. *Nat. Commun.* **11**, 3287 (2020).
18. de Jonge, M. M., Daemen, J., Loriaux, J. M., Steinmann, Z. J. & Huijbregts, M. A. Life cycle carbon efficiency of direct air capture systems with strong hydroxide sorbents. *Int. J. Greenh. Gas Control* **80**, 25–31 (2019).
19. Liu, C. M., Sandhu, N. K., McCoy, S. T. & Bergerson, J. A. A life cycle assessment of greenhouse gas emissions from direct air capture and Fischer–Tropsch fuel production. *Sustain. Energy Fuels* **4**, 3129–3142 (2020).
20. Deutz, S. & Bardow, A. Life-cycle assessment of an industrial direct air capture process based on temperature-vacuum swing adsorption. *Nat. Energy* **6**, 203–213 (2021).
21. Budinis, S. *Direct Air Capture* (International Energy Agency, 2020).
22. *Negative Emissions Technologies and Reliable Sequestration: a Research Agenda* (National Academies Press, 2019); https://doi.org/10.17226/25259
23. Hertwich, E. G. et al. Integrated life-cycle assessment of electricity-supply scenarios confirms global environmental benefit of low-carbon technologies. *Proc. Natl Acad. Sci. USA* **112**, 6277–6282 (2015).
24. McQueen, N. et al. Cost analysis of direct air capture and sequestration coupled to low-carbon thermal energy in the United States. *Environ. Sci. Technol.* **54**, 7542–7551 (2020).
25. Bahar, H. & Bojek, P. *Concentrating Solar Power (CSP)* (International Energy Agency, 2020).
26. Sandalow, D., Friedmann, J., McCormick, C. & McCoy, S. *Direct Air Capture of Carbon Dioxide* (Innovation for Cool Earth Forum, 2018).
27. Baciocchi, R., Storti, G. & Mazzotti, M. Process design and energy requirements for the capture of carbon dioxide from air. *Chem. Eng. Process.* **45**, 1047–1058 (2006).
28. Singh, B., Strømman, A. H. & Hertwich, E. G. Comparative impact assessment of CCS portfolio: life cycle perspective. *Energy Procedia* **4**, 2486–2493 (2011).
29. Hanssen, S. V. et al. The climate change mitigation potential of bioenergy with carbon capture and storage. *Nat. Clim. Change* **10**, 1023–1029 (2020).
30. Querini, F., Dagostino, S., Morel, S. & Rousseaux, P. Greenhouse gas emissions of electric vehicles associated with wind and photovoltaic electricity. *Energy Procedia* **20**, 391–401 (2012).
31. Kätelhön, A., Meys, R., Deutz, S., Suh, S. & Bardow, A. Climate change mitigation potential of carbon capture and utilization in the chemical industry. *Proc. Natl Acad. Sci. USA* **116**, 11187–11194 (2019).
32. Gibon, T., Arvesen, A. & Hertwich, E. G. Life cycle assessment demonstrates environmental co-benefits and trade-offs of low-carbon electricity supply options. *Renew. Sustain Energy Rev.* **76**, 1283–1290 (2017).
33. *Non-exhaust Particulate Emissions from Road Transport: An Ignored Environmental Policy Challenge* (Organisation for Economic Co-operation and Development, 2020).
34. Heck, V., Gerten, D., Lucht, W. & Popp, A. Biomass-based negative emissions difficult to reconcile with planetary boundaries. *Nat. Clim. Change* **8**, 151–155 (2018).
35. Creutzig, F. et al. Considering sustainability thresholds for BECCS in IPCC and biodiversity assessments. *Glob. Change Biol. Bioenergy* **13**, 510–515 (2021).
36. Gabrielli, P., Gazzani, M. & Mazzotti, M. The role of carbon capture and utilization, carbon capture and storage, and biomass to enable a Net-Zero-CO_2 emissions chemical industry. *Ind. Eng. Chem. Res.* **59**, 7033–7045 (2020).
37. Davis, S. J. et al. Net-zero emissions energy systems. *Science* https://doi.org/10.1126/science.aas9793 (2018).
38. Carton, W., Lund, J. F. & Dooley, K. Undoing equivalence: rethinking carbon accounting for just carbon removal. *Front. Clim.* https://doi.org/10.3389/fclim.2021.664130 (2021).
39. Kearns, J. et al. Developing a consistent database for regional geologic CO_2 storage capacity worldwide. *Energy Procedia* **114**, 4697–4709 (2017).
40. Arvidsson, R. et al. Environmental assessment of emerging technologies: recommendations for prospective LCA. *J. Ind. Ecol.* **22**, 1286–1294 (2018).
41. *ISO 14044:2006 Umweltmanagement—Ökobilanz—Anforderungen und Anleitungen* (International Organization for Standards, 2006).
42. *ISO 14040:2006 Umweltmanagement—Ökobilanz—Grundsätze und Rahmenbedingungen* (International Organization for Standards, 2006).
43. Direct air capture to help reverse climate change. *Climeworks* https://www.climeworks.com/page/co2-removal (2020).

44. von der Assen, N., Voll, P., Peters, M. & Bardow, A. Life cycle assessment of CO_2 capture and utilization: a tutorial review. *Chem. Soc. Rev.* **43**, 7982–7994 (2014).
45. Hauschild, M. Z. et al. Identifying best existing practice for characterization modeling in life cycle impact assessment. *Int. J. Life Cycle Assess.* **18**, 683–697 (2013).
46. Holmes, G. et al. Outdoor prototype results for direct atmospheric capture of carbon dioxide. *Energy Procedia* **37**, 6079–6095 (2013).
47. Pehnt, M. & Henkel, J. Life cycle assessment of carbon dioxide capture and storage from lignite power plants. *Int. J. Greenh. Gas Control* **3**, 49–66 (2009).
48. Steubing, B., Wernet, G., Reinhard, J., Bauer, C. & Moreno-Ruiz, E. The ecoinvent database version 3 (part II): analyzing LCA results and comparison to version 2. *Int. J. Life Cycle Assess.* **21**, 1269–1281 (2016).

Acknowledgements
K.M. was supported by a grant from the Eva Mayr-Stihl Foundation under the PROSET project. We thank G. Realmonte for providing additional references for DAC material consumption.

Author contributions
K.M., S.P. and F.C. co-designed the research and jointly wrote the paper. S.D. compiled the process inventories for the TSA DAC plant and conducted the LCA for this technology. K.M. compiled the process inventories for the HT-Aq DAC, conducted the comparative LCA, prepared the figures and compiled the supplementary material. S.P. developed the 1 Gt scale-up comparison with other technology pathways.

Competing interests
The authors declare no competing interests.

Additional information
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41560-021-00922-6.

Correspondence and requests for materials should be addressed to Kavya Madhu.

Peer review information *Nature Energy* thanks Derrick Carlson and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.